Cultural Blend — Kiva funding-speed research

Updated 2026-11-09. One story across the competition work and the October and November research extensions. Every number here is traced in the claims register.

Finalist, UNSW Marketing Analytics Hackathon Challenge 2026. Cultural Blend (Sophia and Tuan) was one of eight finalist teams selected from 44 registered teams involving 133 students at 19 universities across eight countries. The final took place on 4 September 2026. Official UNSW results.

Bottom line

Kiva loans that will fund slowly can largely be spotted from loan details at posting, so the useful response has three steps: **predict** slow loans, **prioritise** lender-side support for them, and **test** before giving any writing advice. In a retrospective check, a boosted tree on loan details flags slow loans 2.63–4.04 times better than chance and beats a simple historical lookup in every test year. Most individual framing associations are small enough to bound, and text adds only a small, inconsistent forecast gain. What the data cannot show is that rewriting a description makes a loan fund faster, so writing advice waits for a small truthful pilot.

Problem and data

Which narrative choices in Kiva loan descriptions are associated with faster funding, and how do those patterns differ across contexts and periods? The project analysed **1,453,846 loan records from 2016–2025**; six negative durations were excluded, leaving **1,453,840 valid completed durations**, including 1,637 refunded records. Outcomes were time from listing to completed funding and completion within 24 hours.

Two facts shape every claim. First, the share of loans funded within a day fell from 46.0% before the pandemic to 30.3% during it and 30.0% after, a descriptive shift that never recovered. Second, 96.4% of loans were disbursed to borrowers before listing, so platform funding time often reflects a partner replenishing capital rather than a borrower waiting. The data contain only loans that eventually funded, so they cannot say whether a loan funds at all.

Approach: three layers of evidence

Layer	Question	Method
August primary run	Which narrative features are associated with speed, and how certain is that?	OLS on log(1 + days) and a binomial GLM, HC3 versus country-clustered uncertainty, within-region contrasts; chronological 2024–2025 holdout for prediction
October extension	Do the associations survive richer controls, including programme text?	Country-clustered OLS/LPM with and without an unvalidated programme-text proxy, BH-adjusted
November benchmark	How much does text add to a forecast built on loan structure alone?	Matched feature blocks on identical rows, as-of training cohorts, five annual windows 2021–2025, paired month and country resampling

A fourth, exploratory set of checks (8–9 November) asks which associations are small enough to rule out a practically large effect, using equivalence tests on the October coefficients, a corrected wild cluster bootstrap, subgroup refits, a historical lookup baseline and tuned forecasters.

Findings

1. Large row counts did not remove country-level uncertainty. Of 120 terms significant under HC3, 64 lost significance once errors were clustered by country. Urgency language went from $p < 0.001$ to $p = 0.49$ for duration. The two regional family-framing patterns each rest on two countries, so their wider heuristic intervals cross zero.

2. Most framing associations are small, and two are unresolved. With a practical bound of about ±5% of funding time and ±1 percentage point of same-day completion per extra mention per 100 words, family, basic-needs, business, agency and third-person framing are bounded in both October specifications. Urgency and first-person framing are not: their intervals still allow associations of roughly 6–14%. "Not significant" for urgency meant "too uncertain", not "no effect". The bound was set after seeing the estimates, so this is exploratory. Within 10 subgroups of period, loan size, gender and region, a wild cluster bootstrap on the programme-adjusted model confirms family and third-person framing within ±5% of funding time in all 10. Business and agency each hold in 9 of 10. Basic needs, gratitude and sentiment hold in 5, 5 and 7 of 10, so they cannot be confirmed in every group; that does not make them large. Of 360 subgroup tests, 6 pass adjustment and none is urgency or agency, so no group shows a confirmed large framing effect.

3. Almost nothing passes the strictest test. Nine measures, two outcomes and two models make 36 tests: 5 pass the country-clustered test, 4 a wild cluster bootstrap, and 1 after adjusting for multiple testing. That one is third-person framing for funding time in the expanded model (−0.9% in 1 + days per extra mention per 100 words; bootstrap p 0.0009, BH p 0.008), or 49.7 minutes on a typical 2.85-day loan. With programme controls it does not pass (bootstrap p 0.128, BH p 0.426). Loan details sit on another scale: a doubled loan amount goes with +44.6% and a month of term with +8.6% in 1 + days. These are associations, not effects of writing.

4. Text adds a small, inconsistent forecast gain. On 728,640 evaluation records, adding description form, lexical measures and TF-IDF lowered the linear model's error in 5 of 5 years, by 0.26–0.39 days (median 0.29), and shuffled text keeps 42–61% of that gain, so about half is context. A compact tree model gained only 0.001–0.042 days. Fixed trees gain up to about 0.15 days from text; in an exploratory run, tuned trees reach 0.16–0.22 in 2021. Country resampling included zero in 4 of 5 linear comparisons. In 2025, text improved error and probability scores but slightly lowered ranking (AUC 0.893 to 0.888). Richer text may encode purpose, location or programme templates as well as framing.

5. Loan details predict most of the variation, and the flag beats a lookup. The primary boosted tree reached MAE 5.20 days and R^2 0.542 on 278,887 holdout loans; the 24-hour classifier reached ROC AUC 0.900. Flagging the top 10% of predicted-slow loans gives 2.63–4.04× lift on each year's slowest 20%, against 1.97–2.74× for a training-only lookup by country, sector and month. The 95% range of that gap is above zero in every year, with a lowest bound of +0.08 in 2021 when resampling by country. Median within-country precision is 0.49–0.56, against 0.22–0.29 for the lookup. An exploratory run tuned a boosted tree and XGBoost on the year before each test year. On loan details, tuned trees were lowest in 3 of 5 years, by at most 0.06 days, and fixed settings won 2021 and 2022, so tuning did not clearly help; XGBoost

beat the tuned boosted tree in 2 of 5 years. These are retrospective predictions on completed loans, not deployment estimates.

6. Loan purpose tracks funding speed, but purpose is not a lever. Against the average loan, Education runs -68.7% and Clothing +30.2% in 1 + days. In an exploratory 20,000-description sample, sanitary and water themes average 1.5 and 1.8 days; they fund fast, but that speed largely reflects which countries and loan types they are, and their adjusted ranges include zero. On all completed loans, the Sanitation & Hygiene sector is modestly faster (-13.4%) and the toilets activity is not distinct from farming (-5.7%, range -19.7% to +10.7%). These are associations among completed loans.

What this means

For Kiva and its partners, the evidence supports a three-step approach. It does not support country-level writing rules or a claim that editing speeds funding.

Predict. Score each listing's risk of slow funding at posting, from loan details. The flag has only been checked on past loans, so it must be recalibrated and validated by country before use.

Prioritise. Send lender-side support to flagged loans: feature them, offer matching funds, alert lenders who follow the sector, and ask the partner to check the listing. 96.4% of loans reach the borrower before listing, so the wait ties up partner capital. What these levers do to funding speed is unmeasured, so each needs a controlled pilot.

Test. Writing advice has to pass three gates, in order:

1. **Validate the measures.** Dual-code a sample of narratives for clarity, specificity, need and agency, then freeze definitions.
2. **Close the data gaps.** Partner identifiers, all listings including expired ones, and original text versions.
3. **Design one truthful pilot** with partners: a clarity prompt, a control, a practical-effect threshold agreed in advance, and enough power to detect it.

Limits

Observational associations; 48 country clusters with sparse regions; a programme proxy that does not identify partners; completed-funding records only; snapshot rather than listing-time field values; retrospective prediction windows. Loan photos were not analysed, because fetching them at scale needs Kiva's permission. Historical slide claims are superseded; see evidence and corrections. The submitted proposal and competition deck are preserved as historical records.

LinkedIn achievement entry

Title: Finalist — UNSW Marketing Analytics Hackathon Challenge 2026

Organisation: UNSW Business School

Date: September 2026

Selected as one of eight finalist teams from 44 registered teams in the UNSW Marketing Analytics Hackathon Challenge 2026. As part of Cultural Blend with Sophia, analysed 1.45 million Kiva loan records to investigate associations between loan narratives and funding speed. Our work combined text analysis, statistical inference, chronological predictive validation and sensitivity checks, with an emphasis on reproducibility and responsible interpretation.

This is an achievement description for Tuan's profile; it does not assign an unverified individual contribution split or describe finalist recognition as a prize win. Official recognition.